PROJECT PHASE I REPORT

ON

HELIOSIGHT

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to

the APJ Abdul Kalam Technological University

in partial fulfillment of the requirements for the award of the degree

of

Bachelor of Technology

in

Computer Science and Engineering



Department of Computer Science and Engineering
St. Joseph's College of Engineering and Technology, Palai

November : 2024

Declaration

We undersigned hereby declare that the project report on **"HELIOSIGHT"**, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala, is a bonafide work done by us under supervision of **Prof Thushara Sukumar** . This submission represents our ideas in our own words and where ideas or words of others have been included. We have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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CERTIFICATE

This is to certify that the report entitled "**HELIOSIGHT**" submitted by **LINTA JOHN (SJC21CS076), SACHIN PHILIP (SJC21CS103), SHAINE THOMAS (SJC21CS108), THOMAS K BINU (SJC21CS123)** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the project work carried out by them under my guidance and supervision.

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Acknowledgement

The success and final outcome of this project phase I required a lot of guidance and assistance from many people, and we are extremely privileged to have received their support throughout the completion of this project. All that we have accomplished is only possible due to their supervision and assistance, and we are sincerely grateful to them.

We would like to express my respect and gratitude to the management of St. Joseph's College of Engineering and Technology for providing me with the opportunity and platform to work on this project.

A special word of thanks goes to our beloved Principal, **Dr. V. P. Devassia**, for providing invaluable support and necessary facilities to carry out this project.

We are extremely indebted to **Dr. Joby P.P**, Professor & Head of the Department of Computer Science and Engineering, for his valuable suggestions and encouragement throughout the course of this project work.

We would like to express our gratitude to our project coordinator, **Prof. Jikku Thomas**, Assistant Professor in the Department of Computer Science and Engineering, for his valuable suggestions and guidelines during the entire duration of this project. We truly appreciate his contributions and technical support in preparing this report.

Our heartfelt thanks go to our project guide, **Prof. Thushara Sukumar**, Assistant Professor in the Department of Computer Science and Engineering, displayed a keen interest in this project and provided guidance and all the necessary information for developing a robust system.

We are thankful and fortunate enough to have received constant encouragement, support, and guidance from all the staff members of the Department of Computer Science and Engineering. Their assistance played a crucial role in the successful completion of my project phase I work.

Abstract

Heliosight aims to develop an interactive geospatial web portal to estimate rooftop solar energy potential using machine learning and deep learning techniques. The system will automatically extract building footprints from high-resolution satellite imagery, utilizing advanced models like U-Net for accurate segmentation. Users can also upload external images to generate detailed rooftop blueprints, which are analyzed for shadows and optimized solar energy mapping. By integrating geospatial solar radiation data, the portal will calculate potential solar energy production for user-selected areas, providing tailored recommendations for the best panel placement and types to maximize energy generation. Dynamic, interactive visualizations will display detailed power generation estimates. The portal is designed to be user-friendly, scalable, and adaptable to various regions, offering individuals, businesses, and policymakers actionable insights for adopting solar energy, contributing to sustainability, and promoting renewable energy usage at a localized level.

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List of Abbreviations

AHP	Analytic Hierarchy Process
CNN	Convolutional Neural Network
DNI	Direct Normal Irradiance
DSM	Digital Surface Model
ER	Entity Relationship
GHI	Global Horizontal Irradiance
GIS	Geographic Information System
LULC	Land Use and Land Cover
PV	Photo Voltaic
SVM	Support Vector Machine

Chapter 1

Introduction

HelioSight is an interactive web portal designed to address the need for precise rooftop solar potential estimation, a key challenge in promoting widespread solar energy adoption. By harnessing machine learning, specifically the U-Net deep learning model, HelioSight automatically identifies building footprints from high-resolution satellite imagery. This data, combined with geospatial solar radiation data, provides users with accurate insights into their rooftop's annual solar energy potential.

HelioSight delivers actionable recommendations on panel placement, type, and configuration through dynamic visualizations, empowering homeowners to make well-informed decisions about solar adoption. This platform makes renewable energy information accessible and promotes sustainable choices, contributing to environmental goals and supporting a transition to cleaner energy.

1.1 Background

The need for accessible and accurate tools to assess rooftop solar energy potential is more pressing than ever as the world moves toward sustainable energy solutions. Yet, a significant barrier to widespread solar adoption is the lack of precise, user-friendly tools that help individuals evaluate the feasibility of solar installations on their buildings. This gap inspired the development of Heliosight, an interactive web portal designed to provide precise solar potential estimations for rooftops, empowering users to make informed,

confident decisions about renewable energy adoption.

Heliosight is grounded in advanced machine learning techniques, particularly the U-Net deep learning model, which automatically detects building footprints from high-resolution satellite images. By integrating this footprint data with geospatial solar radiation information, Heliosight offers accurate estimates of solar energy potential tailored to each user's unique building. This approach allows users to bypass costly, labor-intensive manual assessments, offering a more streamlined, affordable way to understand solar feasibility.With the detailed analysis, the platform generates a 2D rooftop blueprint that accounts for structural details like shadows, slope, and other characteristics crucial to optimal solar panel placement. Additionally, users can refine their estimations by specifying their preferred solar panel area and type, gaining a clear picture of how to maximize energy generation.

The project was selected because it not only addresses the need for reliable solar energy assessment tools but also aligns with broader environmental goals. By enabling accessible, data-driven decisions on renewable energy, Heliosight supports a future where clean energy is accessible to all, promoting solar adoption and sustainability on a global scale.

1.2 Problem Statement

The adoption of rooftop solar energy is often limited by the absence of accessible and precise tools to evaluate solar potential at an individual building level. Current methods for estimating solar energy potential are either too generalized—providing rough estimates that lack the specificity needed for individual buildings—or too complex, requiring technical knowledge that non-expert users may not possess. This limits the ability of homeowners to make informed decisions on solar installations.Accurate rooftop solar assessments also rely on precise building footprint data, which is challenging to extract from satellite imagery. Integrating these footprints with geospatial solar radiation data adds another layer of complexity, as it demands advanced data processing and modeling to

ensure the energy production estimates are reliable and location-specific. Without such integration, existing solutions may misrepresent energy potential, discouraging adoption and leading to suboptimal investments.

To address these issues, Heliosight—a web-based platform that leverages machine learning and deep learning models to automatically extract building footprints from high-resolution satellite imagery. By combining this data with local solar radiation information, Heliosight provides users with accurate, easy-to-understand estimates of their rooftop's solar potential. This platform simplifies decision-making and actionable insights in a user-friendly format.

1.3 Objectives of the Project

1.3.1 User Authentication

The portal includes a robust user authentication system that allows users to create and manage their accounts securely. Upon visiting the portal, users can easily register by providing necessary information, including their email address and a password. Once registered, users can log in to their accounts using their credentials. The system also supports password management features, including password recovery and reset options, ensuring users can regain access to their accounts if they forget their credentials. Additionally, users can update their profile information and preferences, enabling a personalized experience tailored to their needs.

1.3.2 Building Location Selection and Mapping

The portal features an interactive mapping interface that allows users to efficiently locate their building for solar potential assessment. Users can enter their building's address or geographic location into a search bar, which will display the building on a map. Once located, users can select their specific building, ensuring the system accurately captures

the relevant geographic data needed for analysis. This functionality also allows for zooming and panning on the map, providing users with a visual context of their building's surroundings, which is essential for later stages of the solar potential assessment.

1.3.3 Solar Potential Estimation (Footprint-Based)

Using high-resolution satellite imagery, the system employs a U-Net model to extract the building footprint automatically. This footprint serves as the foundational basis for solar potential calculations. The portal integrates geospatial irradiance data, specifically Global Horizontal Irradiance (GHI) and Direct Normal Irradiance (DNI), corresponding to the selected location. Once the building footprint is established, the portal calculates an initial estimate of the solar energy potential based on the footprint and the irradiance values. Users can refine this estimate further by entering specific details about their intended solar panel installation, such as the area of the panels and their type, allowing for a more precise calculation of potential energy output.

1.3.4 Image-Based Rooftop Blueprint Generation

The portal offers a unique image-based analysis option, enabling users to upload images of their rooftops for more detailed assessments. Upon image upload, the system processes these images to generate a comprehensive 2D blueprint of the rooftop. Using the U-Net model for image segmentation, the system identifies and highlights usable rooftop areas, obstacles (such as chimneys or vents), and any features that may impact solar panel placement. The resulting blueprint overlays this information onto the building footprint, providing users with a clear and detailed view of their rooftop's potential for solar energy generation.

1.3.5 Shadow Analysis and Usable Area Calculation

To ensure accurate solar potential estimates, the portal performs an in-depth shadow analysis based on the building's geographic location, the time of day, and seasonal variations. The system calculates the projected shadows cast by nearby structures and obstacles, allowing it to determine which areas of the rooftop will receive direct sunlight throughout the day. Usable rooftop areas for solar panels are then identified by excluding these shadowed regions, providing users with a clear understanding of where they can install panels without obstruction. This detailed analysis enhances the accuracy of energy production forecasts and helps users make informed decisions about their solar investments.

1.3.6 Optimal Panel Placement and Configuration Suggestions

Based on the results of the rooftop blueprint and shadow analysis, the portal provides automated recommendations for optimal solar panel placement. The system takes into account various factors, such as the slope and orientation of the roof, usable area availability, and shading patterns. It suggests the best layout for solar panels to maximize exposure to sunlight, improving overall energy efficiency. Additionally, the portal offers insights into suitable panel types and configurations, including tilt angles and orientations, tailored to the specific characteristics of the rooftop, ensuring users can achieve maximum energy production.

1.3.7 Solar Energy Output Estimation and Summary Report

After completing the analysis and receiving input from the user regarding panel specifications, the portal generates a comprehensive report summarizing the estimated solar potential and expected energy output. This report includes detailed information on the recommended panel placements, configurations, and an overview of the rooftop's usability for solar energy generation. Users can download this report in various formats, such as PDF or CSV, for future reference, allowing them to share insights with stakeholders or use the data for planning their solar panel installations.

1.3.8 Administrative Functions (for Admin Users)

The portal includes essential administrative functions for authorized personnel to manage users and system data. Admins log in securely and, once authenticated, can oversee user account activities, including approving registrations, handling inquiries, and managing account statuses.

Admins are responsible for data integrity, ensuring that irradiance values, building footprints, and other critical data are accurate and current. They can modify existing records, add new data sources, and update system parameters to optimize performance. Additionally, admins can adjust U-Net model configurations for improved building footprint and rooftop blueprint accuracy. After modifications, they can confirm changes and monitor system stability, ensuring the portal functions effectively and remains user-friendly.

1.4 Scope and Limitations

1.4.1 Scope

The Heliosight aims to provide users with a precise, accessible platform to estimate rooftop solar energy potential, targeting homeowners. By leveraging deep learning for building footprint extraction and integrating solar radiation data, the platform offers accurate, location-specific solar potential assessments. Users can interactively explore two analysis pathways: a quick estimate based on footprint data or a detailed blueprint that evaluates shadows, slope, and panel placement. This tool is designed to streamline solar energy decision-making and support sustainable practices on a broad scale, encouraging greater adoption of solar technology.

1.4.2 Limitations

Heliosight's effectiveness is influenced by several limitations. Firstly, the accuracy of building footprint extraction relies heavily on the quality of satellite imagery; poor image resolution can lead to incorrect estimates of solar potential. Additionally, solar irradiance data may vary significantly by geographic location and season, impacting the reliability of energy predictions. Complex rooftop designs, such as those with irregular shapes or extensive shading from adjacent structures, can also complicate the assessment process and result in suboptimal recommendations for solar panel placement. Scalability is another concern; high user volumes during peak times may strain the system, leading to slower processing times and limited access to real-time data. Lastly, while Heliosight strives to be user-friendly, some users may still struggle to interpret the results effectively, especially those without a background in solar energy, which could hinder their decision-making process regarding solar adoption.

1.5 Project Overview

Heliosight is a web-based platform designed to provide users with accurate estimates of rooftop solar energy potential by leveraging machine learning techniques for building footprint extraction from satellite imagery. The platform aims to empower homeowners with reliable insights into solar energy feasibility, facilitating informed decision-making regarding renewable energy adoption. By using advanced deep learning models, Heliosight automatically extracts building footprints and integrates them with geospatial solar radiation data, enabling precise, location-specific energy production estimates. Users can choose between two assessment pathways: a quick estimate based on the building's footprint or a detailed analysis utilizing uploaded rooftop images to create a 2D blueprint. This blueprint evaluates critical factors such as shadows and slope, helping to optimize solar panel placement for maximum energy generation.Ultimately, Heliosight seeks to simplify the solar energy decision-making process, making renewable energy insights more accessible and encouraging sustainable practices across various sectors.

Chapter 2

Literature Review

2.1 Extracting Building Footprints from Satellite Images using Convolutional Neural Networks,2018

The paper[1] details a CNN-based method for accurately detecting building footprints from high-resolution satellite images. This process involves an encoder-decoder CNN architecture inspired by U-Net, designed to generate pixel-wise segmentation maps that identify buildings with high precision despite variations in building shapes, sizes, and textures.

The approach begins with pre-processing, where 512x512 RGB images undergo normalization to standardize intensity values and improve the model's pattern recognition. Data augmentation techniques further enhance training robustness by increasing dataset variability. The CNN architecture itself consists of contracting and expanding paths: the contracting path extracts features at multiple levels, while the expanding path upsamples these features to restore the spatial resolution. Skip-connections between these paths help the network retain fine details in the segmentation. Following segmentation, post-processing algorithms refine the results, using geometric techniques like the Douglas-Peucker algorithm to simplify and smooth building boundaries, thereby reducing noise and enhancing the visual clarity of detected structures.

While the method achieves impressive precision, with object-based evaluations yielding 99% precision and 100% recall, its accuracy depends heavily on the quality and diversity of the training dataset. The model's performance may be limited when applied to regions with building features that differ significantly from those in the training data. Additionally, complex urban landscapes with dense, irregularly shaped buildings can challenge segmentation accuracy, sometimes resulting in merged or fragmented structures. Lastly, the reliance on RGB satellite images means it may struggle to differentiate buildings from other structures in regions with minimal contrast or intricate layouts.

2.2 Assessment of Solar Energy Potential using GIS and AHP approach: A Case Study of Bumthang Valley,2020

The paper[2] employs the Analytic Hierarchy Process (AHP) integrated with Geographic Information System (GIS) to identify optimal locations for solar photovoltaic (PV) installations in Bhutan's Bumthang Valley. This approach uses multiple weighted criteria—including solar irradiance, slope, elevation, aspect, land use and land cover (LULC), proximity to roads and settlements, and exclusions for protected areas and biological corridors. The AHP technique assigns weights to each factor, which are then used in GIS to generate a suitability map, identifying potential locations for solar energy infrastructure. The study further calculates annual solar energy potential for these identified areas, based on varying PV efficiencies.

Despite its robust methodology, the study acknowledges some limitations. Key factors like local humidity, temperature, and proximity to transmission lines were excluded due to data constraints, potentially affecting site suitability accuracy. Additionally, the solar irradiance data relied upon may lack precision due to the absence of long-term, ground-based measurements. These limitations make the findings provisional, with recommendations for further studies to incorporate these missing factors to obtain more comprehensive results.

2.3 Analysis and Positioning of 2D Solar Panel,2018

The paper[3] involves a systematic approach to assess solar panel placement using a 2D mapping system. The proposed system uses data from Google Maps to calculate the optimal positioning of solar panels on rooftops, considering factors like sunlight intensity, local weather patterns, and shading from nearby structures. Users input details such as average electricity consumption and sun exposure hours. The system then calculates the number of panels needed to meet energy requirements and creates a 2D visual layout showing how panels would appear on the user's property.

The system's reliance on 2D images restricts it from providing comprehensive spatial details, as it lacks 3D modeling capabilities that could better account for the angles and heights of surrounding structures. The system's dependence on Google Maps data may also limit accuracy, particularly in areas with outdated or low-resolution images. Additionally, the absence of real-time, localized solar irradiance and temperature data may reduce the precision of energy estimates. Future improvements suggested include adding 3D modeling for enhanced visualization and incorporating more dynamic weather and irradiance data sources.

2.4 Remote sensing using drone and machine learning for computation of rooftop solar energy potential,2023

The study[4] leverages remote sensing through drone imagery and machine learning to evaluate rooftop solar energy potential in a rural area. The process begins with high-resolution data collection using a quadcopter drone that captures aerial images, which are then processed into an orthophoto and a digital surface model (DSM). These images, with a spatial resolution of approximately 3 cm, provide a foundation for detailed analysis of rooftop features and allow accurate mapping of available roof space.

Once the orthophoto and DSM are created, a convolutional neural network (CNN), specifically a U-Net model, segments the building roofs from other elements in the images. This model isolates rooftop areas suitable for solar panel installations, allowing precise estimation of available rooftop space. The segmented rooftop areas are then vectorized and processed for further calculations.Shadow removal follows, addressing the shadows cast by nearby buildings and trees that could hinder accurate solar potential estimation. The study models the sun's path throughout the year, taking into account variations in sun azimuth and elevation. By creating shadow masks for different seasonal extremes, only shadow-free portions of the rooftop are considered for energy calculations. This step is crucial to ensure accurate estimates by excluding obstructed areas.

The final stage involves calculating energy potential based on solar irradiance data, incorporating seasonal variations and inclined surface measurements. By using monthly sunshine hours and solar orientation, the model estimates potential energy output for each rooftop, allowing for optimal energy predictions for each season.Despite the method's strengths, it faces certain limitations. Drone-based data collection can be challenging over extensive areas due to limited flight range and battery constraints. The segmentation model may also struggle with highly intricate roof structures, leading to minor inaccuracies in roof area estimation. Additionally, the model assumes consistent solar irradiance; real-world variations in weather conditions could impact actual energy production, making accuracy somewhat variable in changing climates.

2.5 An Approach for Estimating Solar Photovoltaic Potential Based on Rooftop Retrieval from Remote Sensing Images,2018

For estimating[5] rooftop solar photovoltaic (PV) potential begins with collecting high-resolution remote sensing images and digital surface models (DSM) to capture 2D rooftop outlines and 3D parameters like slope and aspect. Using an object-oriented classification approach, rooftops are detected and classified into types—flat, shed, gable, hipped,

and mansard. This classification facilitates calculating PV module parameters, including rooftop area, tilt angle, and orientation. Shadow effects from surrounding structures are considered using sun position data, ensuring accurate calculation of non-shadowed PV areas. Finally, solar PV potential is estimated by calculating the irradiance on these surfaces over time.

However, this approach has limitations. The dependency on high-resolution satellite imagery and DSM data is resource-intensive, which can be challenging for larger areas. Additionally, the classification model may struggle with complex rooftop geometries, impacting the precision of rooftop area extraction. Seasonal and environmental variations, such as cloud cover and pollution, may also affect real-world solar irradiance, making potential estimates somewhat less accurate in dynamic climates.

2.6 Predicting the Solar Potential of Rooftops using Image Segmentation and Structured Data,2021

The paper[6] presents a methodology to estimate the solar potential of building rooftops by combining computer vision, machine learning, and structured data. The approach begins with semantic segmentation of aerial imagery to detect roof sections and objects, using a U-Net architecture with a ResNet-34 backbone. This allows for precise allocation of space available for solar panels. The next step involves estimating the roof's pitch and azimuth: pitch is predicted through a linear model and Random Forest algorithm based on building features, while azimuth is determined geometrically. To account for shading effects, a shading mask is generated using a Sky-View factor technique that factors in both surrounding buildings and topographic features. The annual solar potential is computed by combining these geometric parameters with meteorological data to assess potential photovoltaic output.

A primary limitation of this methodology is the lack of validation datasets, particularly for the shading algorithm, which limits full accuracy verification across the pipeline. Instead, each algorithmic step is individually validated, but the entire pipeline's results

are largely verified through comparison with client-provided data in the energy sector. Additionally, roof object segmentation and pitch prediction show lower accuracy due to limited labeled data and the inherent difficulty of inferring pitch from aerial images alone. These limitations may impact accuracy but are mitigated by factors like azimuth and surface area, which have greater influence on solar potential.

2.7 Building Rooftop Extraction Using Machine Learning Algorithms for Solar Photovoltaic Potential Estimation,2023

The paper[7] presents a two-stage methodology to estimate the solar photovoltaic (PV) potential of building rooftops using machine learning algorithms and spatial analysis tools. The first stage focuses on rooftop detection from satellite images, employing pre-processing steps like gamma correction, shadow masking, and noise filtering, followed by machine learning models (Support Vector Machine and Naïve Bayes) for rooftop extraction. This stage achieved high accuracy, with SVM reaching an F1-score of 94.7% and Naïve Bayes achieving 92.2%. In the second stage, the estimated rooftop areas are used in PV potential modeling using tools like PVWatts, PVGIS, and ArcGIS, which incorporate solar irradiance data to estimate energy production.

The study found that PVGIS, which uses African data, provided the most accurate estimation of 24.9 GWh/year for the study area. A primary limitation is the dependence on high-resolution images with consistent spectral properties for effective segmentation and accurate rooftop detection. Additionally, the lack of diverse regional datasets can affect the model's accuracy for broader applications. Improvements could be achieved with higher-quality imagery from advanced satellites or drones, which would enhance detection precision and reduce errors in energy estimation.

2.8 Footprint Extraction

This project uses a U-Net deep learning model to detect buildings figure 2.1 in satellite images, aiming to automate building identification for applications like urban planning, disaster response, and renewable energy assessment. The process involves loading satellite images and labeled building masks, pre-processing them into numpy arrays, and training the U-Net model to predict building locations by learning from ground-truth data. After training, the model's predictions are compared with actual building masks, with yellow regions in the images representing detected buildings.

While this approach provides accurate building segmentation, it has several drawbacks. First, the model's accuracy depends heavily on the quality and diversity of the training data; poor or biased datasets can lead to inaccurate predictions or generalization issues in different geographic areas. Second, the model may struggle with complex landscapes, dense urban areas, or varying building shapes, resulting in missed or incomplete detections. Shadows, vegetation, and adjacent structures can also interfere with predictions.

Additionally, the computational resources required for training and processing high-resolution imagery can be significant, making this method costly and time-intensive. Finally, fine-tuning is often necessary to adapt the model for specific applications, which can be challenging and may require advanced knowledge in deep learning.

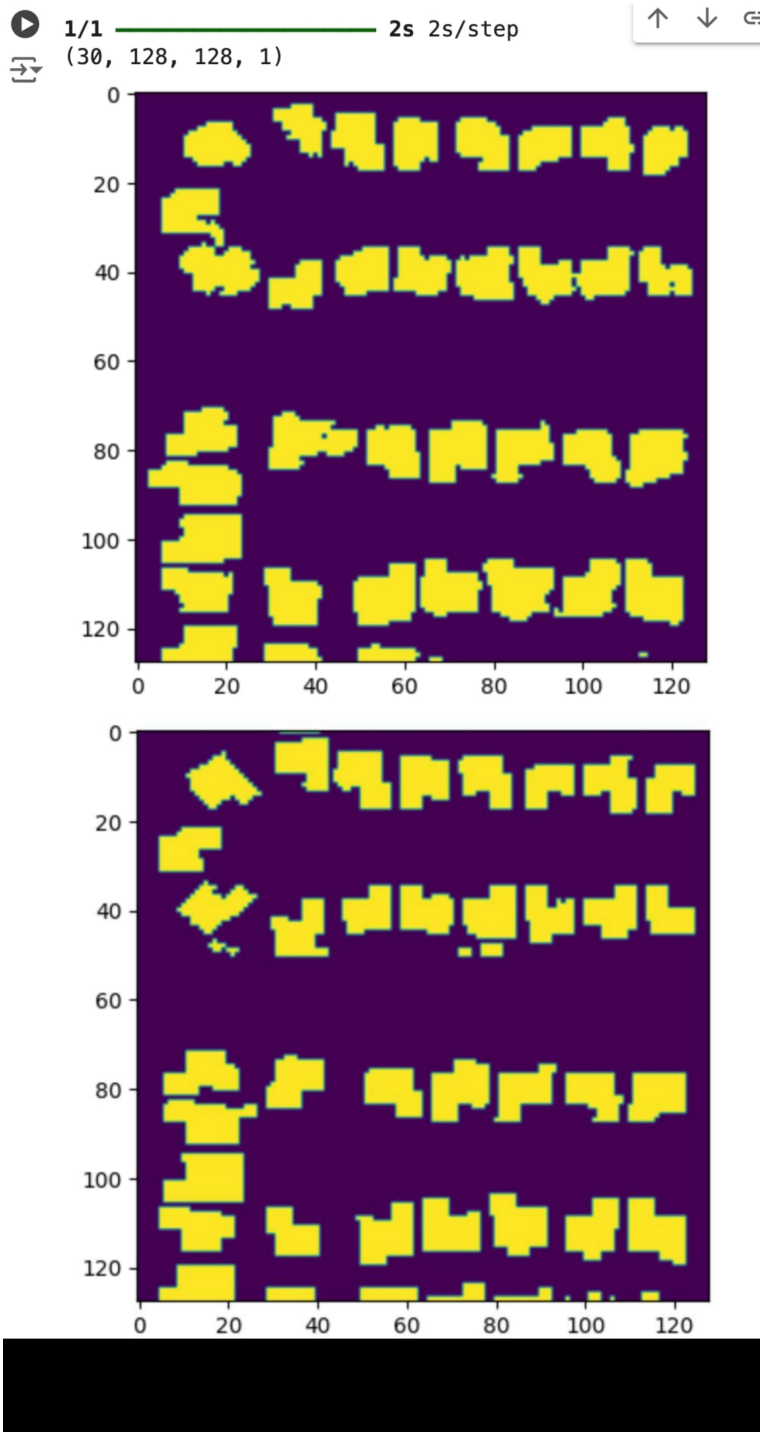


Figure 2.1: Building Footprint

Chapter 3

Requirement Analysis

3.1 Software Requirements

3.1.1 Functional Requirements

Footprint Detection and Rooftop Segmentation

1. The system shall allow users to upload rooftop images in supported formats (e.g., JPEG, PNG).
2. The system shall process images using the U-Net model, segmenting rooftop areas and identifying usable versus obstructed zones.
3. The system shall store processed segmentation results in the database for retrieval and display.
4. The system shall display segmented images with color-coded zones to indicate usable and obstructed areas.
5. The system shall allow users to re-upload or replace images and re-run segmentation.

2D Blueprint Generation

1. The system shall use vectorization tools (e.g., QGIS, Shapely) to generate a 2D blueprint based on segmented rooftop data.
2. The system shall store generated blueprints in the database, enabling easy retrieval and integration with other data layers.
3. The system shall display blueprints in a user-friendly format, with clear labels for usable areas, obstacles, and rooftop boundaries.
4. The system shall provide download options for blueprints in common formats (e.g., PDF, PNG).

Shadow and Sun Exposure Analysis

1. The system shall use pvlib-python to calculate solar irradiance, sun position, and shading patterns based on geographic location and time.
2. The system shall provide a visualization overlay of shadow patterns on the 2D blueprint, adjustable by season and time of day.
3. The system shall store irradiance data and shadow analysis results in the database for repeated user access.
4. The system shall allow users to view and interact with seasonal shading patterns, enabling informed solar panel positioning.

Energy Production Estimation

1. The system shall calculate energy production potential based on rooftop area and irradiance data retrieved from external sources.
2. The system shall display energy estimates as daily, monthly, and annual projections, with visual graphs and charts for user interpretation.

3. The system shall store energy production results in the database, associated with each rooftop analysis.

4. The system shall allow users to view and compare energy estimates under different scenarios, such as varying panel coverage.

User Authentication and Profile Management

1. The system shall support user registration with an email and password, with password recovery and reset options.

2. The system shall provide secure login and session management to protect user data.

3. The system shall allow users to manage profile information, such as email, password, and preferences.

4. The system shall allow users to view their past analyses and retrieve previously generated blueprints, segmentation results, and energy estimates.

Admin Management Console

1. The system shall allow admins to view and manage registered user accounts, including deactivation or reactivation options.

2. The system shall provide an analytics dashboard displaying key metrics on usage trends, popular features, and system performance.

3. The system shall allow admins to review system error logs and diagnostic reports to identify issues and optimize performance.

4. The system shall allow admins to manage stored data, including the ability to delete outdated or erroneous analysis results and maintain data integrity.

3.1.2 Non-Functional Requirements

Performance Requirements

- **Response Time**: The system shall deliver responses within 2-3 seconds for standard database queries and user interactions. Computational tasks such as rooftop segmentation and shadow analysis should process within 25-40 seconds for typical image sizes (up to 5 MB).
- **Scalability**: The system shall be able to handle at least 1,000 concurrent users without a significant decrease in performance, with the capacity to scale as needed.
- **Availability**: The system should maintain an uptime of at least 99.5%, ensuring high availability for users accessing the platform for solar assessment.
- **Data Throughput**: Data requests, including image uploads and retrievals, shall be processed with a minimum bandwidth requirement of 10 Mbps to support large image handling.

Usability Requirements

- **User Interface**: The user interface shall be intuitive and accessible, catering to both general users and admin users. It should support responsive design principles for easy use on desktops, tablets, and mobile devices.
- **User Documentation**: A comprehensive user guide with sections on setup, key features, and troubleshooting shall be provided, along with tooltips and inline assistance within the platform.
- **Learnability**: New users shall be able to perform basic tasks (e.g., uploading images, viewing results) within 10 minutes of their first interaction with the platform.

Reliability Requirements

- **Fault Tolerance**: The system shall be able to recover from failures, such as a cloud function timeout, and retry or notify the user without data loss.
- **Backup and Recovery**: Supabase and other storage components shall support automated data backup every 24 hours and ensure recovery within 2 hours of a critical failure.
- **Data Consistency**: All user interactions, data updates, and computational outputs shall ensure transactional integrity, maintaining a consistent state across the database.

Security Requirements

- **Data Encryption**: All sensitive data, including user credentials and images, shall be encrypted both in transit and at rest.
- **Data Privacy**: User data, particularly geospatial and energy estimation data, shall not be shared with third parties unless explicitly permitted by the user.

Maintainability Requirements

- **Code Modularity**: The codebase shall be modular, allowing individual components (e.g., segmentation model, database integration) to be updated independently.
- **Documentation**: All code and APIs shall be documented to support future maintenance and development, including architectural and database schema documentation.
- **Monitoring and Logging**: The system shall implement error logging, user activity tracking, and performance monitoring, ensuring quick identification and resolution of issues.

Portability Requirements

- **Cloud Provider Independence**: The system should be designed to allow migration from one cloud provider to another (e.g., from AWS to Google Cloud) with minimal reconfiguration.
- **Browser Compatibility**: The platform shall be compatible with modern browsers, including Chrome, Firefox, Safari, and Edge, and maintain consistent performance across all supported devices.
- **Device Compatibility**: The platform shall function effectively on various device types, including desktops, tablets, and smartphones.

3.2 Heliosight: The Proposed Solution

Heliosight is an innovative web-based solution designed to empower homeowners with an accurate assessment of rooftop solar energy potential. By leveraging advanced image processing, geospatial analysis, and machine learning, HelioSight aims to simplify and streamline the solar assessment process. Users can upload rooftop images, input specific location details, and receive both quick estimates and detailed analyses on potential solar power generation for their property.At the core of Heliosight's technical architecture is a U-Net model for image segmentation, enabling precise detection of building footprints and rooftop features directly from user-uploaded images. This model identifies usable areas, accounting for obstructions like vents or chimneys, to offer optimized panel placement recommendations. To further enhance blueprint generation, Heliosight employs vectorization techniques using tools like shapely and QGIS, which turn segmented images into vector polygons, creating a 2D blueprint of usable rooftop zones.

Additionally, Heliosight integrates solar irradiance and shadow analysis through libraries like pvlib-python, which calculates solar angles, irradiance, and potential shading based on the property's location and surrounding structures. This analysis is essential in providing accurate energy production estimates by identifying optimal sun exposure

throughout the year.Heliosight is built on a robust backend, leveraging Supabase as the primary data storage, authentication, and API layer, ensuring secure user management and efficient data retrieval. Compute-intensive tasks, such as image segmentation and solar analysis, are processed on cloud GPU-enabled instances, allowing for high-speed performance and scalability. Supabase's PostGIS extension enables storage and management of geospatial data, including building footprints, irradiance information, and vectorized blueprints.

By providing easy-to-understand visualizations and reports, Heliosight offers users actionable insights into solar energy adoption. This solution supports sustainable energy goals by making solar assessments accessible, accurate, and user-friendly, helping users take meaningful steps.

Chapter 4

System Design

4.1 Activity Diagram

4.1.1 Admin Activity Diagram

The figure 4.1 outlines the workflow for an admin managing a portal system. The process starts with the admin logging into the portal and proceeding to authenticate their credentials. If the credentials are validated successfully, the admin is granted access to the portal's system management functionalities. However, if the validation fails, an error message is displayed, and access is denied.

Upon accessing the management functions, the admin has two primary options: updating model configurations or updating database records. If the admin chooses to update model configurations, they can adjust the parameters of the U-Net model, which is likely used for building footprint or solar energy analysis. Following parameter adjustments, the admin can reconfigure settings and deploy updates to apply the changes. The system then prompts the admin to confirm the adjustments, ensuring that the modifications have been successfully implemented.

Alternatively, if the admin opts to update database records, they can modify or add new data within the system's databases. This includes tasks like updating irradiance

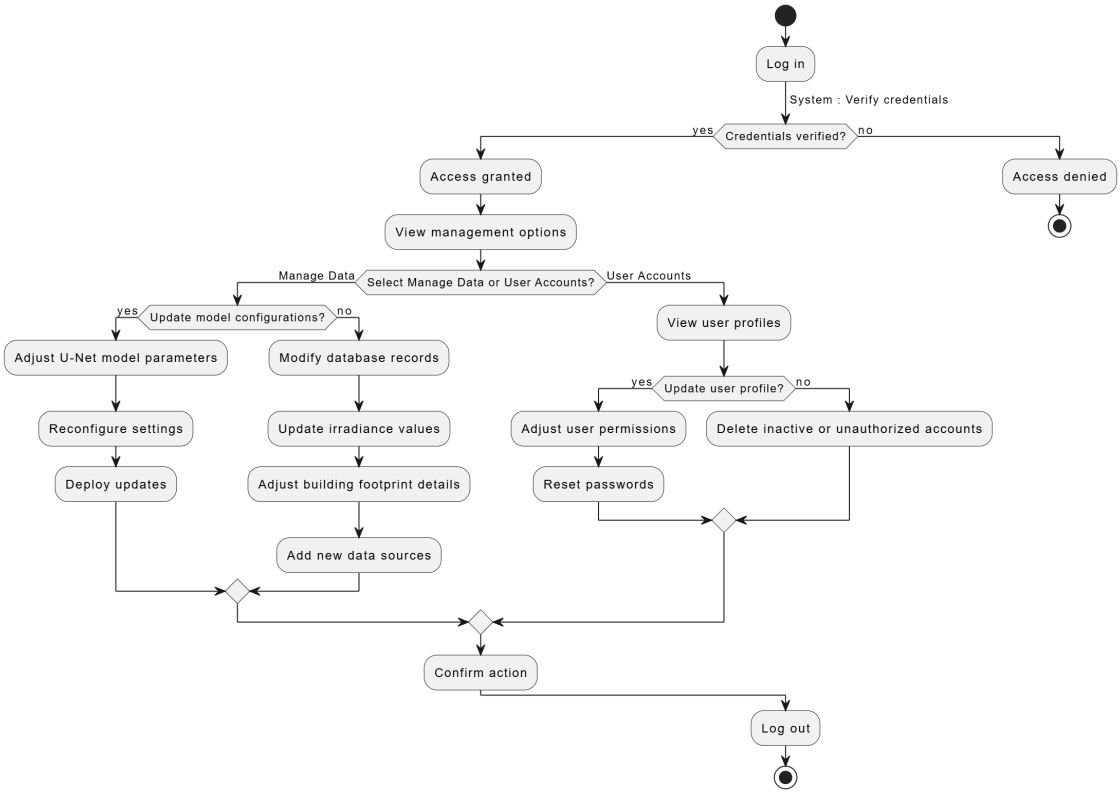


Figure 4.1: Admin Activity diagram

values (crucial for solar energy calculations), adjusting building footprint details to reflect any structural changes, and adding new data sources to expand the dataset available for analysis. After completing these updates, the system prompts for confirmation of successful modifications to verify that all changes have been recorded accurately.

If the admin opts to manage user accounts, they can view user profiles and either update user profiles or delete inactive or unauthorized accounts. Updating a user profile may involve adjusting user permissions or resetting passwords to maintain security and proper access levels. After completing any of these actions, the admin confirms the changes and logs out to end the session securely. This structured workflow ensures that the system's data and user accounts are effectively managed and maintained.

Finally, once the admin completes the chosen tasks, they can log out of the system, closing their session and ensuring secure access management. This workflow provides a structured approach for maintaining and updating critical aspects of the portal system.

4.1.2 User Activity Diagram

The figure 4.2 presents a detailed process for estimating rooftop solar energy potential through an interactive online portal, starting with user authentication. Initially, the user logs into the portal, where the system verifies their credentials. If the validation is unsuccessful, an error message is displayed, stopping further progression. However, if the login is successful, the user is taken to the next phase, where an interactive map is displayed. Here, the user can enter the location of their building and select the specific building they want to analyze. The system then retrieves key data about the building's footprint, which is essential for the initial solar potential calculation.

Following this, the user is given two options for proceeding with the estimation. They can either opt for a quick, initial estimation based on the retrieved building footprint and irradiance data, or they can choose to upload rooftop images for a more accurate, customized estimate. If the user chooses the initial estimation option, the system uses the footprint and irradiance data to generate a basic estimate of the building's solar potential, which is then displayed to the user. At this stage, the user can either accept the initial estimate if it meets their needs or decide to seek a more refined estimate by uploading rooftop images.

If the user proceeds with the image upload, the system processes these images to generate a detailed rooftop blueprint. This blueprint provides a precise representation of the rooftop's structure and dimensions, which is then combined with irradiance data to produce an enhanced solar potential estimate. This refined estimate accounts for shadows, roof angles, and other specific features that the footprint alone might not capture, thus improving accuracy.

Next, the user is asked to input specific details about the solar panel they intend to install, including the panel's area and type. These parameters allow the system to adjust the energy estimation based on the actual specifications of the equipment to be used, enabling a more realistic forecast of energy production.

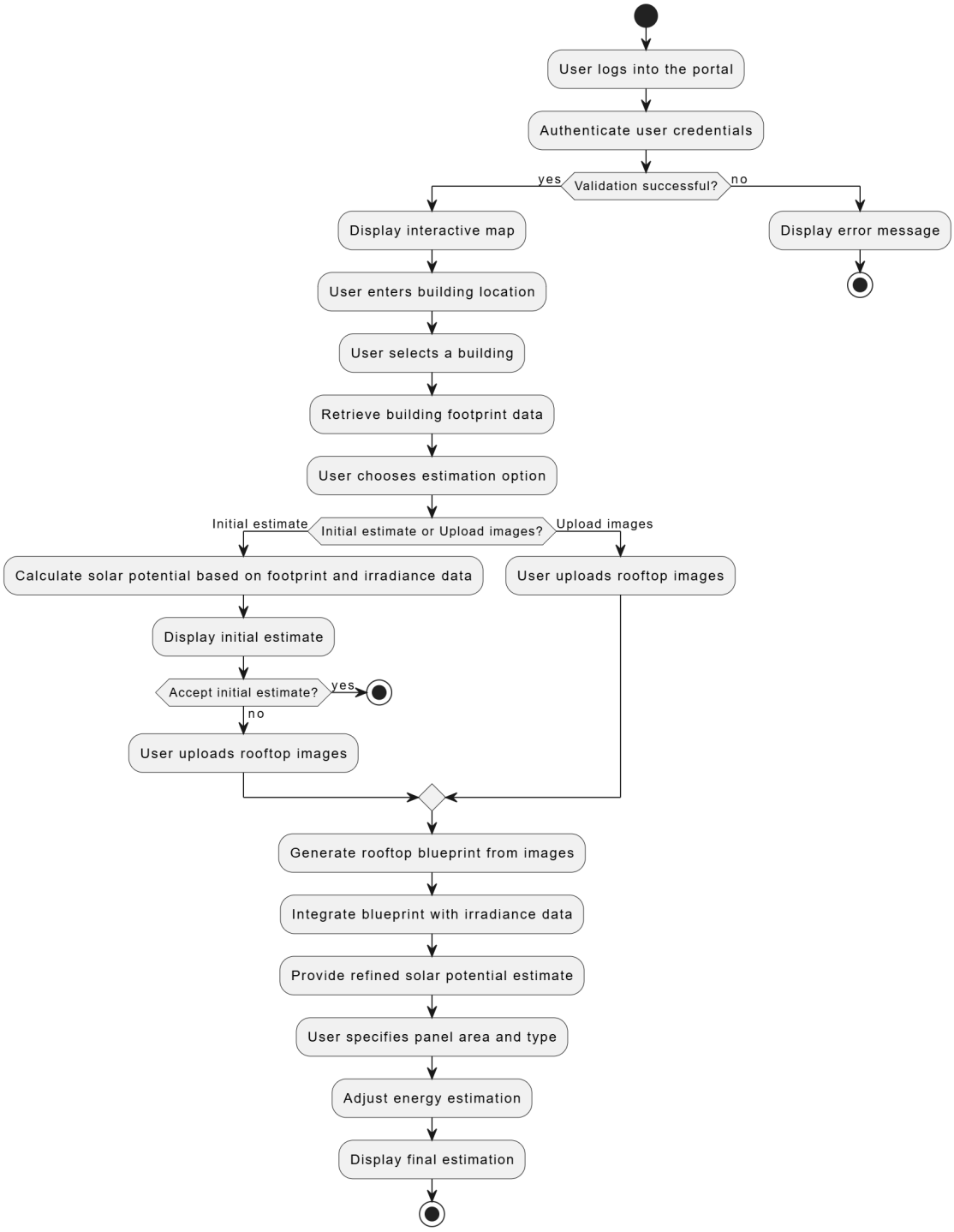


Figure 4.2: User Activity diagram

Finally, with all the information integrated, the system provides a comprehensive, final energy estimation, reflecting the estimated solar potential based on the user's building, rooftop specifics, and selected solar panel attributes. This final estimate gives the user a reliable prediction of the energy output they can expect, allowing for informed decision-making on solar panel installation and energy management.

4.2 ER Diagram

Figure 4.3 showcase the Entity Relationship(ER) diagram represents a data model for a rooftop solar energy estimation system, showing key entities and their relationships. The main entities in this model are User,Building,Blueprint, AnalysisResult, and SolarRadiationData.

The User entity captures information about each user, including attributes like ‘userId‘, ‘name‘, ‘email‘, and ‘role‘. Each user can own multiple buildings, as represented by the ”owns” relationship between the User and Building entities.

The Building entity stores details about each building, including ‘buildingId‘, ‘location‘, ‘footPrintPath‘, ‘createdDate‘, ‘userId‘ (the owner), ‘roofType‘, ‘area‘, and ‘imagePath‘. Each building is owned by a single user, and users can own multiple buildings. Additionally, each building can use multiple sets of solar radiation data from the SolarRadiationData entity, allowing for detailed solar energy calculations. The ”uses” relationship represents this association between Building and SolarRadiationData.

The Blueprint entity represents the architectural plans associated with each building. It includes attributes like ‘blueprintId‘, ‘blueprintPath‘, ‘blueprintDate‘, ‘buildingId‘, ‘shadowDetected‘, and ‘roofSlope‘. Each building has a single associated blueprint, connected through the ”represents” relationship. The blueprint provides essential information, such as the roof slope and shadow detection, which is critical for accurate solar energy estimation.

The AnalysisResult entity captures the output of the solar energy analysis for each building. Attributes include ‘resultId‘, ‘suggestedPanelType‘, ‘calculationDate‘, ‘optimalPanelPlacement‘, and ‘estimatedEnergy‘. Each building has a single analysis result, represented by the ”has” relationship with the AnalysisResult entity, which provides insights into the suggested solar panel type, optimal placement, and estimated energy output.

Lastly, the SolarRadiationData entity holds data on solar radiation measurements, with attributes like ‘dataId‘, ‘location‘, ‘GHI‘ (Global Horizontal Irradiance), ‘DNI‘ (Direct Normal Irradiance), ‘date‘, and ‘time‘. This entity allows the system to access his-

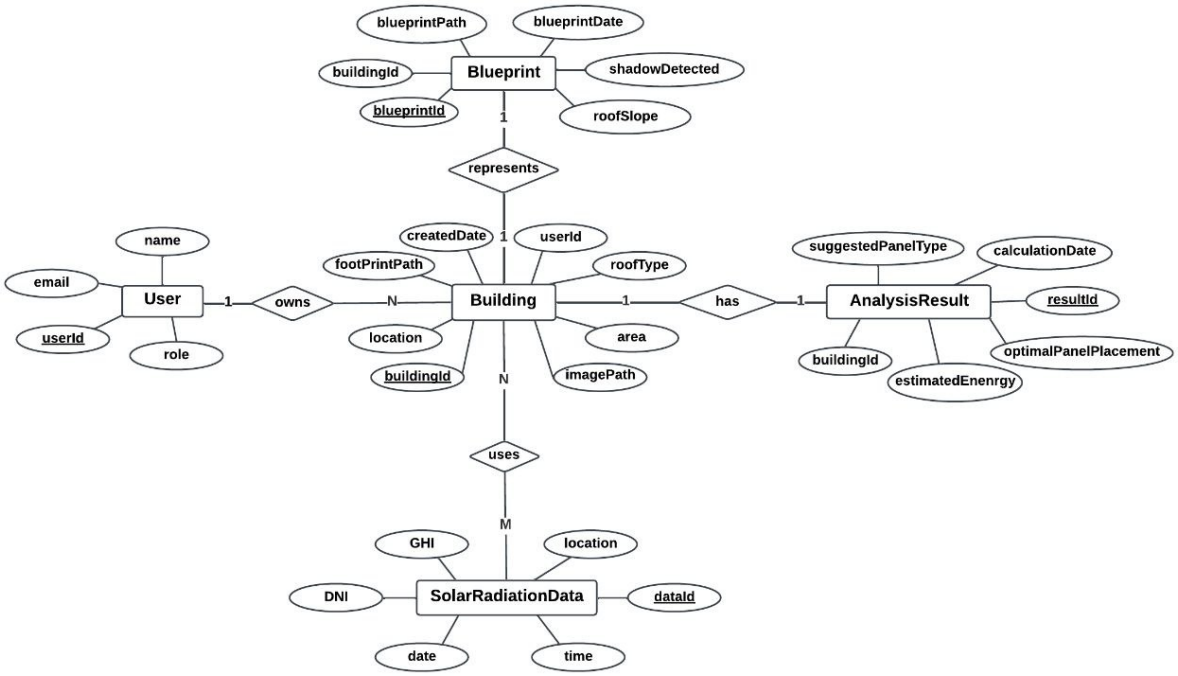


Figure 4.3: ER diagram

torical solar data specific to a location, which is used in energy calculations for the building. The "uses" relationship between Building and SolarRadiationData is many-to-many, meaning each building can utilize multiple sets of solar radiation data for analysis.

In summary, this model provides a structured way to manage user information, building details, architectural plans, solar radiation data, and energy analysis results, supporting accurate rooftop solar energy potential estimation.

4.3 Sequence Diagram

The figure 4.4 depicts a comprehensive process for evaluating the solar energy potential of a building's rooftop using user input, image analysis, irradiance data, and administrative controls. The workflow begins with the user initiating the process by registering an account, where they provide essential details like name, email, and password. The system validates these details, ensuring that the information meets the necessary criteria, and then securely stores the user data in the database. Once registered, the user proceeds to log in by submitting their credentials. The system checks these credentials against the stored information in the database and confirms a successful login, granting the user access to the platform.

Upon logging in, the user is presented with a map interface, allowing them to navigate and select a building for solar potential analysis. When the user selects a building and chooses the "Solar Potential" option, the system begins the assessment process by generating a building footprint—a digital outline of the selected structure. To ensure accurate solar potential calculations, the system requests irradiance data, specifically Global Horizontal Irradiance (GHI) and Direct Normal Irradiance (DNI) values, from the database. This data represents the solar radiation available at the location and is crucial for estimating potential solar energy production. Using this data, the system performs an initial calculation of the solar potential for the selected building, providing the user with preliminary energy output estimates.

After displaying these initial estimates, the system prompts the user to upload rooftop images to enhance the analysis. These images allow for a detailed assessment of the rooftop area, including segment identification and shadow analysis. Once the user uploads the images, the system processes them using image analysis algorithms, identifying segments, analyzing shadows, and creating a precise rooftop blueprint. If necessary, the system may retrieve updated irradiance data to refine the solar potential estimate further, ensuring that the calculations account for the most recent conditions. With this refined data, the system recalculates the solar potential, providing an updated and more accurate energy estimate. It then displays a refined estimate along with tailored recommendations for

solar panel installation, indicating the optimal placement and number of panels based on the rooftop's characteristics.

In addition to the user's interaction, the diagram includes administrative functions. The admin has the capability to manage user information, with options to update or delete user profiles as needed. These administrative actions are essential for maintaining a secure and organized database, ensuring that only authorized users have access. Once any user management actions are completed, the system performs data integrity checks to safeguard the database from inconsistencies or errors. This sequence provides a thorough process for determining solar panel suitability, integrating user inputs, detailed rooftop analysis, solar irradiance data, and administrative oversight to deliver accurate and actionable recommendations for solar energy production.

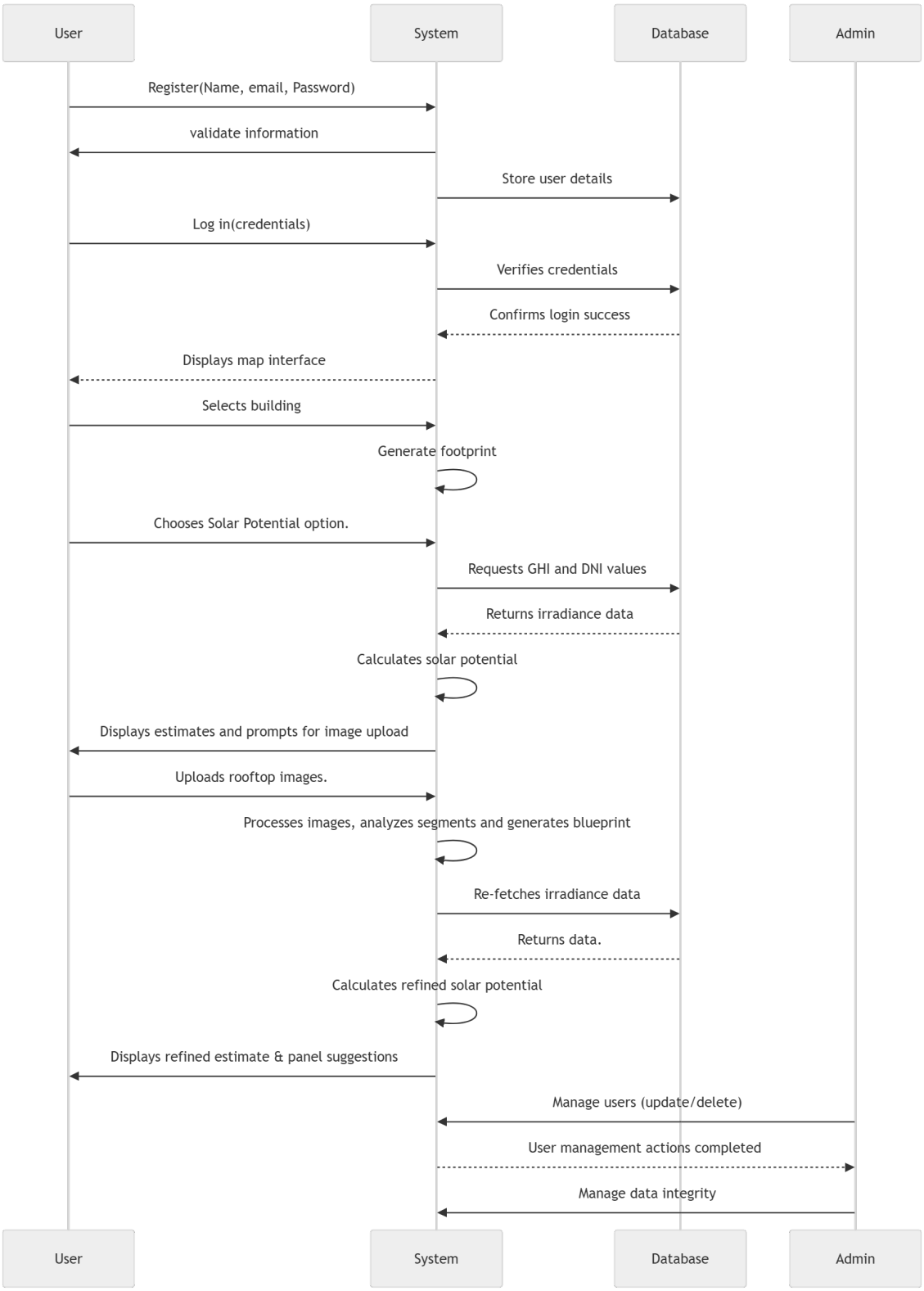


Figure 4.4: Sequence Diagram

4.4 Use Case Diagram

The figure 4.5 illustrates The Roof Solar Energy Estimation Portal is a tool designed to help users estimate the solar energy potential of their buildings using advanced geospatial and machine learning technologies. This interactive platform serves both regular users and administrators, offering a range of functionalities to aid in detailed solar potential assessment.

The user begins by selecting their building's location on an interactive map. This selection allows the portal to gather essential geospatial data, including the building's footprint and solar radiation levels (such as Global Horizontal Irradiance and Direct Normal Irradiance) in that area. By leveraging this data, users receive an initial estimate of the potential solar energy output based on their building's footprint.For users seeking more accuracy, the portal provides an option to upload rooftop images of their building. These images are then processed by deep learning models, like U-Net, to generate a precise rooftop blueprint. This blueprint highlights areas that are suitable for solar panel placement, taking into account factors like shadows and surface characteristics that could affect solar exposure.

After generating the rooftop blueprint, the portal offers a refined solar potential calculation. This refined estimation considers the specific characteristics of the rooftop and user-provided details such as panel type and area. It provides users with a more tailored assessment of expected energy generation and gives recommendations for optimal solar panel placement and configuration, aiming to maximize energy output.Users can also view the solar potential results, which include details on estimated energy generation and cost savings. This information can guide individuals, businesses, and policymakers in making informed decisions on solar energy adoption, contributing to sustainable energy goals.

Administrators, on the other hand, have access to additional functionalities. They can manage user accounts, ensuring smooth operation and addressing any access-related issues. They are also responsible for managing data integrity within the portal, overseeing the accuracy and security of data to maintain reliable outputs and user trust.

Overall, the Roof Solar Energy Estimation Portal provides a comprehensive, data-driven approach to evaluating and optimizing rooftop solar potential, encouraging widespread adoption of renewable energy solutions.

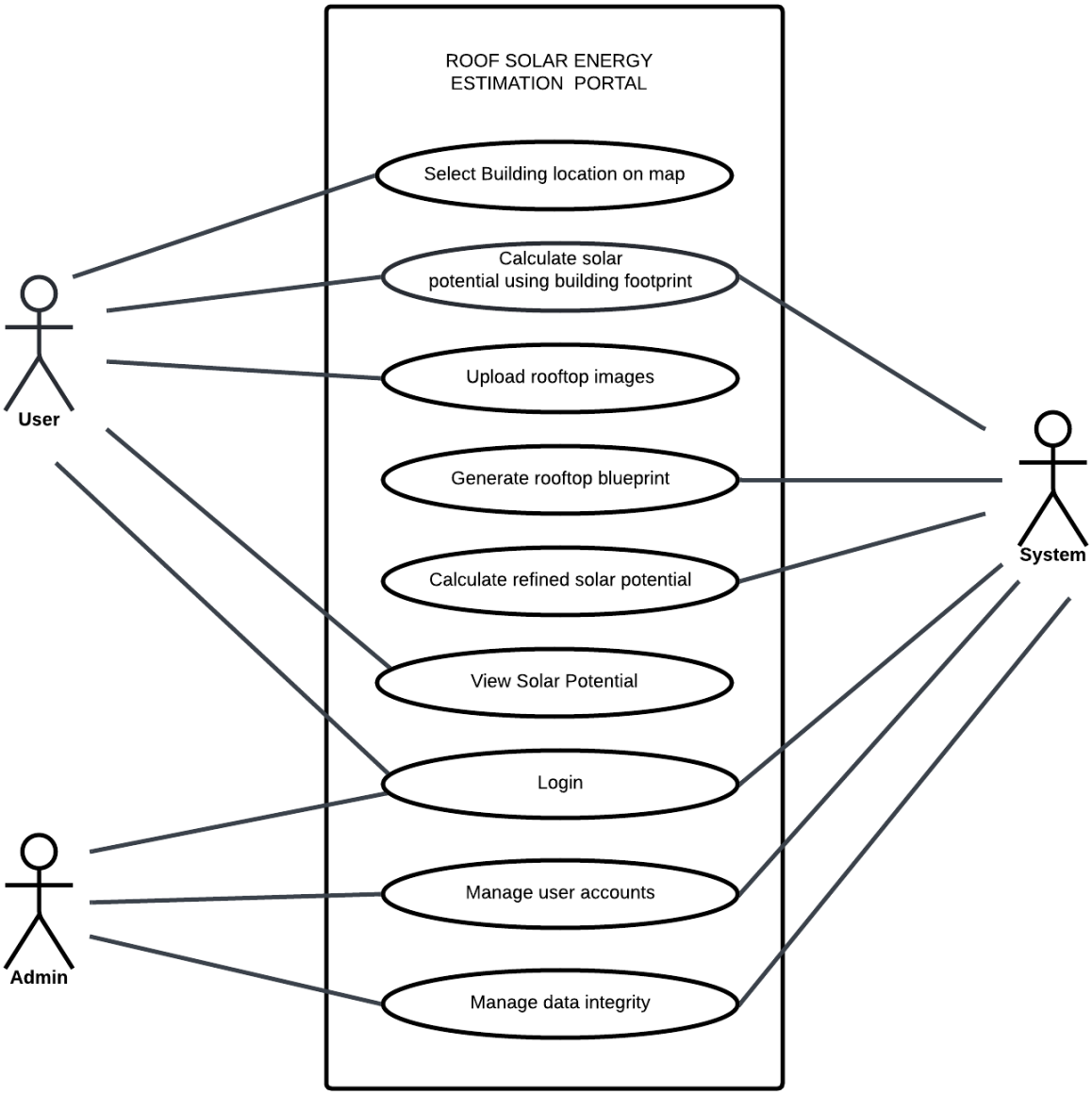


Figure 4.5: Use case diagram

Chapter 5

Conclusion

The HelioSight project offers a straightforward and accessible solution for anyone interested in understanding the solar energy potential of their rooftop. By combining modern technology like machine learning, geospatial analysis, and cloud computing, HelioSight provides reliable solar assessments based on precise data and detailed analysis. Users can either receive quick estimates based on basic information or upload rooftop images for a more in-depth evaluation that includes 2D blueprint generation, shading analysis, and optimal panel placement.

HelioSight is designed with ease of use in mind. Users can interact with a responsive web interface to access visual insights, energy production estimates, and recommendations tailored to their specific rooftop. By automating complex processes behind the scenes, the portal gives users a seamless experience without requiring any technical expertise.

This project supports environmental sustainability by enabling individuals and businesses to make informed decisions about solar energy. HelioSight not only helps users assess solar viability but also encourages greater adoption of renewable energy by removing the guesswork from solar planning. In doing so, HelioSight makes solar energy assessment accessible to a wider audience, contributing to greener communities and a more sustainable future.

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[2] Automatic Building Footprint Extraction using Deep Learning , Jai G Singla , Bhavya Ramani,2023 International Conference on Computational Intelligence, Communication Technology and Networking (CICTN).

[3] Assessment of Solar Energy Potential using GIS and AHP approach, A Case Study of Bumthang Valley , Ugyen Tempa , Jai Govind Singh.2020 International Conference on Electrical and Electronics Engineering (ICE3-2020).

[4] Analysis and Positioning of 2D Solar PanelSanjay Lohar, Ridhesh Hirpara, Tushar Kalsara , Sharvari Patil,2018 International Conference on Smart City and Emerging Technology (ICSCET).

[5] Remote sensing using drone and machine learning for computation of rooftop solar energy potential , Prakash P. S. , Pavan R. Vyas , 2023, Irish Centre for High End Computing (ICHEC), University of Galway (UoG), Ireland , Visvesvaraya Technological University (VTU), Karnataka, India.

[6] An Approach for Estimating Solar Photovoltaic Potential Based on Rooftop Retrieval from Remote Sensing Images,2018.

[7] Predicting the Solar Potential of Rooftops using Image Segmentation and Structured Data,2021.

[8] Bhttps://ieeexplore.ieee.org/

[9] https://isro.hack2skill.com/2024/

ANNEXURE

CSD 415
MAJOR PROJECT
FINAL REVIEW

Team Guide:

Prof. Thushara Sukumar

Team Members:

Sachin Philip (SJC21CS103)
Shaine Thomas(SJC21CS108)
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HELIOSIGHT

ABSTRACT

This project develops an interactive web portal for estimating rooftop solar energy potential using machine learning models like U-Net for building footprint extraction. Users can upload images to generate rooftop blueprints, analyze shadows, and optimize solar energy mapping. The system provides recommendations for solar panel placement and energy estimates through dynamic visualizations. With a user-friendly interface and spatial database integration, the portal offers accurate insights for efficient solar energy adoption.

INTRODUCTION

- The widespread adoption of solar energy is often hindered by the lack of precise tools for estimating rooftop solar potential.
- This project addresses this issue by developing an interactive geospatial web portal that accurately estimates rooftop solar energy potential.
- Utilizing machine learning and deep learning models like U-Net, the system automatically extracts building footprints from satellite imagery and integrates geospatial solar radiation data.
- The platform provides detailed, dynamic visualizations of potential solar energy production, empowering users to make informed decisions on solar adoption.

PROBLEM STATEMENT

- **Lack of Accessible Tools:** There is a shortage of accessible and accurate tools for estimating the solar energy potential of building rooftops at a localized level.
- **Challenges in Building Footprint Extraction:** Obtaining precise building footprint data from satellite imagery presents a significant challenge.
- **Difficulty in Data Integration:** Integrating building footprint data with solar radiation information for meaningful energy production calculations is complex.
- **Current Methods' Limitations:** Existing methods are either too generalized for individual building assessments or too complex for non-experts to use effectively.
- **Need for an Automated Solution:** There is a need for an automated, user-friendly, and scalable solution that can extract building footprints, estimate solar energy potential, and present the information interactively for informed decision-making.

PROPOSED SOLUTION

- **Interactive Web Portal:** The solution involves developing an interactive web portal to estimate rooftop solar energy potential.
- **Data Collection and Preprocessing:** It includes collecting high-resolution satellite images and solar radiation data, preprocessing the images, and using machine learning models like U-Net for accurate building footprint extraction.
- **Solar Energy Production Calculation:** The portal will calculate potential solar energy production based on these footprints and radiation data.
- **User Interaction:** Users can interact with dynamic visualizations and maps to view energy estimates.
- **User-Friendly Interface and Backend Integration:** The system will feature a user-friendly interface and backend integration with a spatial database, aiming to provide actionable insights for adopting solar energy with ease of use and accuracy.

LITERATURE SURVEY

TITLE	METHODOLOGY	CONTENT	LIMITATIONS
[1]Extracting Building Footprints from Satellite Images using Convolutional Neural Networks,2018	Uses a Convolutional Neural Network (CNN) with a U-Net architecture to extract building footprints from satellite images. Pre-processing normalizes images, and post-processing refines building boundaries for improved accuracy.	An approach for extracting and detecting building footprints from satellite images. The proposed technique is capable of extracting buildings having different shapes, sizes and textures.	Potential reduced accuracy and generalizability when applied to satellite images with varying quality, resolution, and building appearances beyond the dataset it was trained on.

TITLE	METHODOLOGY	CONTENT	LIMITATIONS
[2]Assessment of Solar Energy Potential using GIS and AHP approach: A Case Study of Bumthang Valley.2020	Uses in this study involves assessing solar energy potential using a geographic information system (GIS) and the Analytic Hierarchy Process (AHP). Multiple criteria, including solar irradiance, elevation, slope, and proximity factors, were weighted using AHP to generate a suitability map.	The paper assesses solar energy potential in Bumthang Valley, Bhutan, using GIS and AHP. It identifies suitable locations for solar PV installations, finding 6.54 km^2 of land capable of generating over 1000 GWh/year, promoting renewable energy diversification in Bhutan.	Exclusion of factors like humidity, temperature, proximity to transmission lines, and financial analysis, as well as reliance on provisional solar irradiance data lacking long-term ground measurements.
[3]Analysis and Positioning of 2D Solar Panel,2018	The methodology uses Google Maps data and user inputs to calculate solar panel needs, provides a 2D visual layout of installations, estimates energy savings, and connects users with local solar vendors.	This system assesses solar energy potential for buildings by modeling environmental factors, calculating solar panel requirements, and providing visual representations and vendor information for optimal solar panel installation.	The system's accuracy depends on the quality of input data from sources like Google Earth and historical weather patterns, and may be limited by the availability and resolution of such data.

TITLE	METHODOLOGY	CONTENT	LIMITATIONS
[4]Remote sensing using drone and machine learning for computation of rooftop solar energy potential,2023	Using drone-acquired images to create orthophotos and digital surface models. A convolutional neural network model segments building roofs, calculates rooftop solar energy potential, and removes shadows to estimate solar power generation.	This study uses drones and machine learning to compute rooftop solar energy potential by generating orthophotos, digital surface models, and removing shadows, enabling more accurate solar energy estimates.	The study's estimates exclude building roof slopes and directions, which could refine accuracy. It also does not account for diffuse radiation or extreme seasonal weather impacts on energy potential.
[5] An Approach for Estimating Solar Photovoltaic Potential Based on Rooftop Retrieval from Remote Sensing Images,2018	The study uses object-oriented classification to extract 2D rooftop outlines, recognizes 3D rooftop features (slope, area), and combines these with solar radiation data to estimate rooftop solar PV potential and energy generation.	The paper develops a method to estimate rooftop solar photovoltaic potential by analyzing 2D and 3D rooftop features from remote sensing images and DSM, applied in Beijing's Chaoyang District for energy planning.	Limitations include rooftop shape complexity, shading effects, data resolution constraints, and potential inaccuracies in remote sensing classification. These challenges hinder the precision and scalability of the method for large-area solar assessments.

TITLE	METHODOLOGY	CONTENT	LIMITATIONS
[6] Predicting the Solar Potential of Rooftops using Image Segmentation and Structured Data,2021	The approach uses aerial image segmentation via U-Net, predicts roof pitch and azimuth using Random Forest and geometric algorithms, and estimates solar potential by combining roof geometry, meteorological data, and shading effects.	The paper presents a method to estimate rooftop solar energy potential using image segmentation and machine learning, focusing on urban areas, to predict electricity production based on roof geometry and solar exposure.	Challenges include limited labeled data for roof objects, resulting in low segmentation accuracy, and a lack of validation datasets for shading calculations, affecting the pipeline's overall accuracy and comprehensive validation.
[7] Building Rooftop Extraction Using Machine Learning Algorithms for Solar Photovoltaic Potential Estimation,2023	The study applies image pre-processing techniques (gamma correction, shadow masking, segmentation) and machine learning algorithms (SVM, Naïve Bayes) for rooftop extraction, followed by solar potential estimation using PVWatts, PVGIS, and ArcGIS..	The paper estimates rooftop solar photovoltaic potential using machine learning algorithms like SVM and Naïve Bayes for rooftop detection from satellite images, applied in Egypt's Madinaty City for renewable energy generation.	Challenges include varying precision in rooftop detection across different regions, reliance on accurate satellite imagery, and potential discrepancies in solar potential estimation between software tools, impacting prediction reliability and accuracy..

FUNCTIONAL REQUIREMENTS

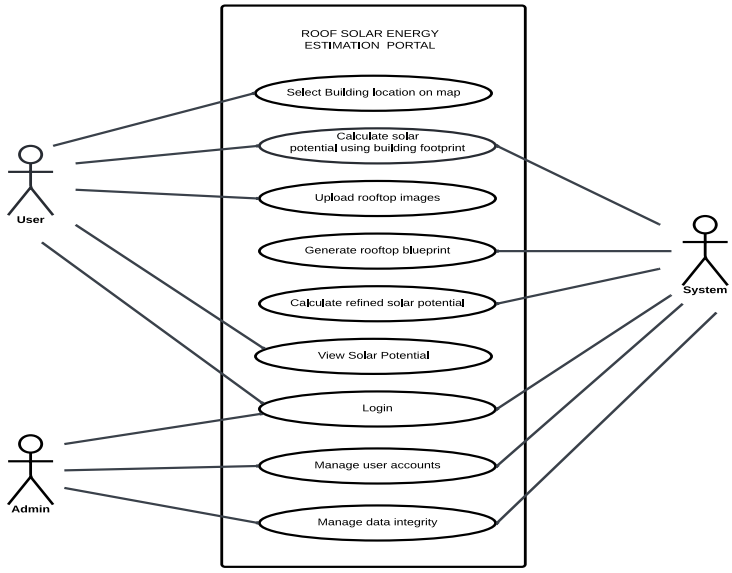
- **Building Footprint Extraction:** Extract building footprints from satellite imagery using U-Net
- **Image Upload:** Allow users to upload images for rooftop blueprints
- **Solar Radiation Integration:** Integrate solar radiation data for energy estimation
- **Panel Recommendations:** Provide optimized solar panel placement suggestions
- **Dynamic Visualization:** Display interactive and detailed power generation estimates
- **User Authentication:** Implement user login and profile management.

NON FUNCTIONAL REQUIREMENTS

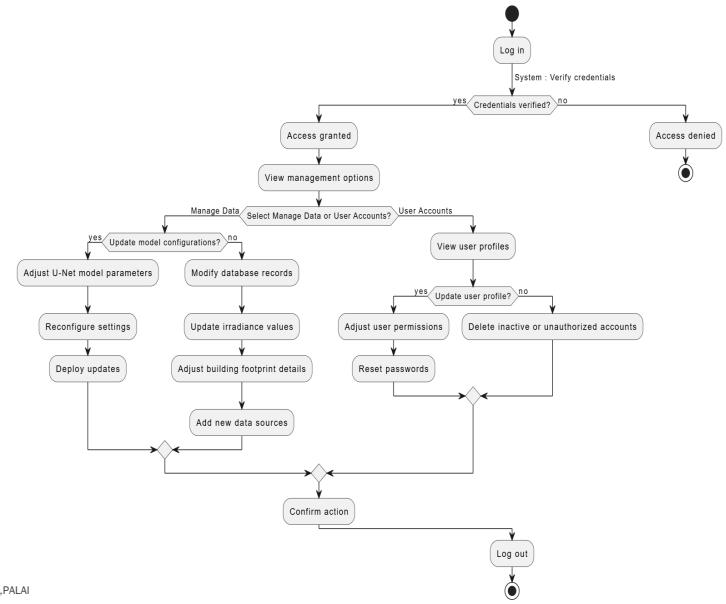
- **Usability:** Ensure the interface is intuitive and easy to use
- **Performance:** Process data quickly and accurately
- **Reliability:** Maintain high availability and handle errors effectively
- **Security:** Protect user data with encryption and access controls
- **Scalability:** Handle large data volumes and user traffic efficiently
- **Compliance:** Adhere to data privacy and security regulations.

DESIGN DIAGRAMS

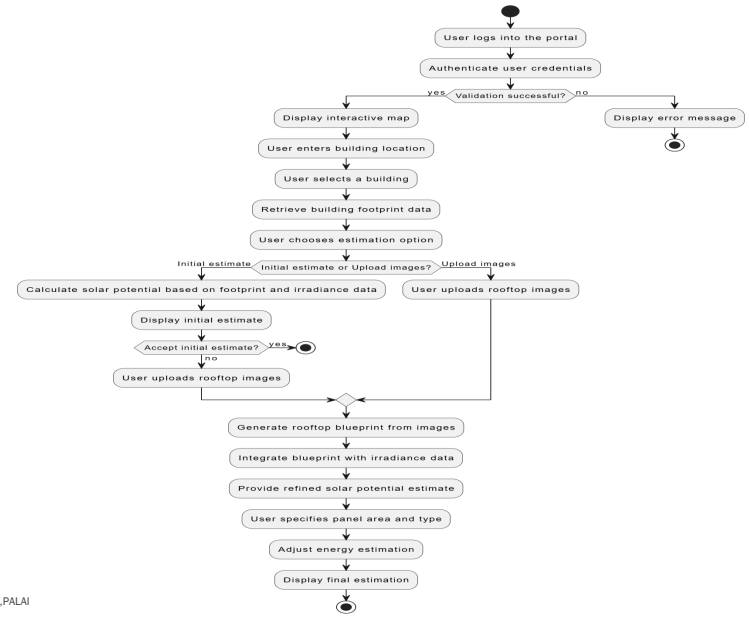
USE CASE DIAGRAM



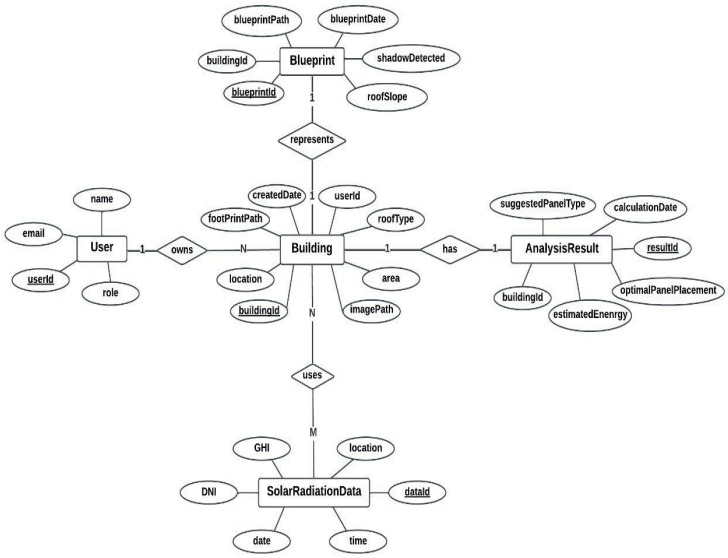
ADMIN ACTIVITY DIAGRAM



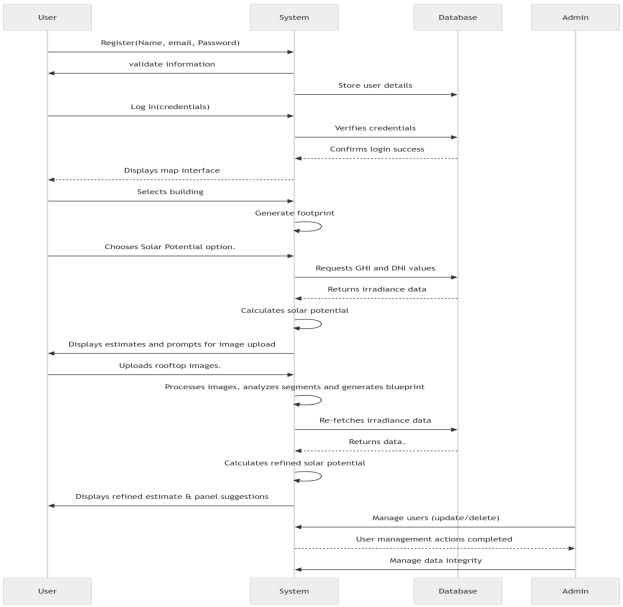
USER ACTIVITY DIAGRAM



E R DIAGRAM



SEQUENCE DIAGRAM



METHODOLOGY

- Collect and preprocess satellite imagery and solar radiation data for the target regions
- Utilize a U-Net model to accurately extract building footprints from the satellite imagery
- Using deep learning model U-Net and Mask R-CNN for roof segmentation from satellite and user-uploaded images, followed by blueprint generation, shadow and slope analysis, and solar panel optimization based on geospatial solar radiation data
- Integrate building footprint data with solar radiation information to estimate potential solar energy production and provide panel placement recommendations
- Create a user-friendly web portal with dynamic visualizations to present energy estimates and recommendations interactively

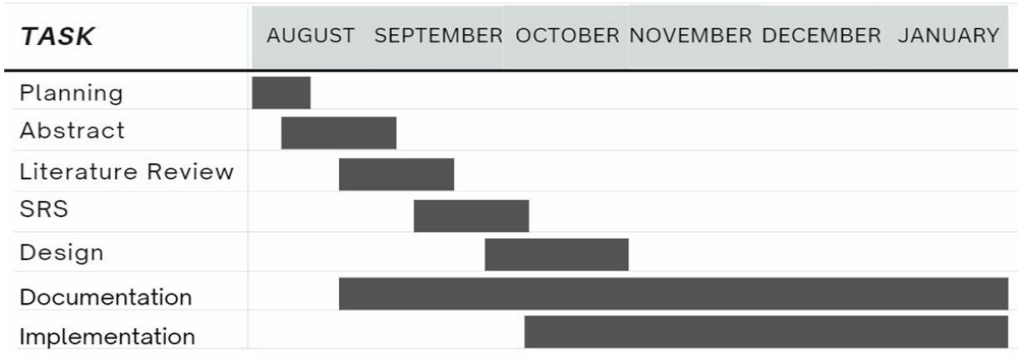
MODULE DESCRIPTION

1) **User Authentication :** Manages user login, registration, and secure access
2) **Building Footprint Extraction :** Detects and extracts building boundaries from satellite imagery
3) **Energy Potential Estimation :** Calculates rooftop solar potential using extracted data
4) **Admin Management :** Oversees user activities, datasets, and system access
5) **Panel Positioning Module :** Calculates optimal panel placement based on rooftop area and orientation

SOFTWARE REQUIREMENTS

- **Database:** Supabase's PostgreSQL
- **Frontend:** React.js
- **Backend:**.Node.js
- **Machine Learning (Model Hosting & Serving):** TensorFlow/Keras(Python)

GANTT CHART



CONCLUSION

- This project delivers a comprehensive, interactive web portal for estimating rooftop solar energy potential by leveraging advanced machine learning techniques.
- It provides users with precise, actionable insights through dynamic visualizations and an intuitive interface.
- By enabling accurate solar energy assessments, the system supports informed decision-making for individuals, businesses, and policymakers, promoting broader adoption of renewable energy and contributing to sustainability goals.

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- [3] Analysis and Positioning of 2D Solar PanelSanjay Lohar, Ridhesh Hirpara, Tushar Kalsara , Sharvari Patil,2018 International Conference on Smart City and Emerging Technology (ICSCET)
- [4]Remote sensing using drone and machine learning for computation of rooftop solar energy potential , Prakash P. S. , Pavan R. Vyas , 2023, Irish Centre for High End Computing (ICHEC), University of Galway (UoG), Ireland , Visvesvaraya Technological University (VTU), Karnataka, India
- [5]An Approach for Estimating Solar Photovoltaic Potential Based on Rooftop Retrieval from Remote Sensing Images,2018
- [6]Predicting the Solar Potential of Rooftops using Image Segmentation and Structured Data,2021
- [7]Building Rooftop Extraction Using Machine Learning Algorithms for Solar Photovoltaic Potential Estimation,2023

THANK YOU....

VISION & MISSION OF THE DEPARTMENT

VISION

- To evolve as a school of computing with globally reputed Centre's of excellence and serve the changing needs of the industry and society.

MISSION

- The department is committed in bringing out career-oriented graduates who are industry ready through innovative practices of teaching and learning process
- To nurture professional approach, leadership qualities and moral values to the graduates by organizing various programs periodically
- To acquire self-sustainability and serve the society through research and consultancy